MAJOR PROJECT TECHNICAL REPORT

MeMOTR Vehicle Tracking Enhancement

Scene-Adaptive Detection Threshold (C2) and Perspective-Weighted Loss (C1) for Multi-Object Tracking in Adverse Conditions

UA-DETRAC · VisDrone · IOTD (Pending) | Phases 1–6 Interim | September 2026

47	62.13	5+5	209
Test Sequences	Baseline HOTA	Training Epochs	C2 Params

1 Project Overview

MeMOTR	C1 + C2	50.4 M	209
Base architecture — end-to-end transformer MOT	Two novel contributions to MeMOTR	Frozen backbone parameters	Trainable ThresholdAdapter params (C2)

This project extends **MeMOTR** (Memory-augmented Transformer for Multi-Object Tracking) with two targeted contributions designed to improve vehicle detection and tracking in **adverse lighting conditions**, specifically the night-time and dusk scenarios present in the Indian On-road Traffic Dataset (IOTD).

MeMOTR uses track-query embeddings that propagate across frames, giving it strong long-term association capability. However, its fixed detection threshold ($\tau = 0.5$) is ill-suited for scenes where genuine vehicles score 0.35–0.45 (dark/low-contrast). The MeMOTR backbone and encoder are **not modified** — all contributions attach as lightweight modules, preserving pretrained generalisation.

> **DESIGN CONSTRAINT**
>
> All contributions operate behind feature flags (use_c1, use_c2) defaulting to False, ensuring zero regression for the unmodified baseline.

2 Problem Statement

Issue 1 — Fixed Detection Threshold (→ C2)

MeMOTR's runtime tracker admits new detections only when classification score $\geq \tau = 0.5$. In dark/dusk IOTD scenes, genuine vehicles frequently score 0.35–0.45, causing silent false-negatives. In bright overexposed conditions, the same threshold admits background clutter.

Issue 2 — Perspective-Blind Training Loss (→ C1)

Uniform loss weight λ across all frame regions fails to account for perspective geometry. In overhead footage, vehicles at varying distances appear at very different scales; a perspective-weighted $\lambda(x,y)$ would better calibrate the model for multi-scale scenarios.

3 Novel Contributions

C2 — ThresholdAdapter (Scene-Adaptive Detection Threshold)

Status: ■ Implemented and Evaluated

Hypothesis: The detection admission threshold τ should adapt per-frame to scene conditions. In dark scenes a lower τ recovers genuine vehicles; in bright/cluttered scenes a higher τ suppresses false positives.

SceneAnalyzer (analytic, 0 learnable params) computes a 3-D feature vector:

- Brightness — mean pixel intensity [0, 1]
- Contrast — standard deviation of intensity [0, 1]
- Track density — active track count / max_tracks [0, 1]

ThresholdAdapter — MLP 3 → 16 → 8 → 1, then sigmoid-and-scale:

```
τ = τ_min + (τ_max − τ_min) × sigmoid(MLP(scene_vec)) = 0.3 + 0.4 × sigmoid(…) ∈ [0.3, 0.7]
```

Total trainable parameters: **209**. MeMOTR backbone remains fully frozen (0 of 50.4 M params trainable) throughout C2 training. Training supervision uses brightness-based proxy labels (MSE loss between predicted τ and brightness-derived target τ).

C1 — PositionMLP (Perspective-Weighted Training Loss)

Status: ■ Architecture ready — training deferred pending IOTD annotation

A small MLP that maps each ground-truth box's (x, y) image position to a per-box loss weight λ(x, y). Boxes near the camera (typically near the image bottom) receive higher λ. Boxes near the horizon receive lower λ. Requires real perspective calibration from IOTD.

4 Datasets

DATASET	SPLIT	SEQUENCES	FORMAT	STATUS
UA-DETRAC	train + test	40 test	MOT Challenge	■ Active
VisDrone Vehicles	train + val	7 val	VisDrone (filtered)	■ Active
IOTD	all	TBD (≥3)	Custom	■ Pending

VehicleMOT — a custom unified dataset class was built to handle mixed-format image naming (UA-DETRAC uses img%05d.jpg, VisDrone uses %07d.jpg) via auto-detection, without renaming files. Designed with sequence-level splits only, pickle-safe module-level factory (Windows spawn), batch size 1, num_workers=0.

5 Implementation Phases

#	Phase	Status	Key Deliverables
1	**Environment Setup & Baseline Verification**	■ Complete	Compiled MSDA CUDA op; patched TrackEval Python 3.9 issues; smoke test zero-regression verified; hardcoded paths fixed; git submodules added.
2	**Data Pipeline**	■ Complete	VehicleMOT dataset class; UA-DETRAC XML→MOT conversion; VisDrone vehicle category filtering; seqinfo.ini generation for all 47 sequences.
3	**Module Development**	■ Complete	SceneAnalyzer, ThresholdAdapter (C2) implemented and wired into RuntimeTracker behind use_c2=False flag. PositionMLP (C1) architecture designed.
4	**IOTD Integration**	■ Deferred	Requires ≥3 annotated IOTD sequences. Perspective calibration for C1 blocked on this.
5	**Interim Baseline Training**	■ Complete	5 epochs on UA-DETRAC+VisDrone; loss 21.4→2.2; checkpoints 0–4 saved; baseline HOTA = 62.13.
6	**C2 Training & Interim Evaluation**	■ Complete (Interim)	ThresholdAdapter trained 5 epochs (MSE 0.001465→0.000687); C2 HOTA = 60.42; clamp verified; failure mode diagnosed.
6+	**Full C1+C2 Training**	■ Planned	After IOTD annotation: joint C1+C2 training with real night calibration.

6 Phase 5 — Baseline Training

5 epochs
Training duration

2e-5
Learning rate

21.4→2.2
Loss trajectory

0
Backbone params trained (frozen)

EPOCH	AVG LOSS	CHECKPOINT
0	~21.4	checkpoint_0.pth
1	~8.1	checkpoint_1.pth
2	~4.8	checkpoint_2.pth
3	~3.1	checkpoint_3.pth
4 (eval)	~2.2	checkpoint_4.pth

7 Phase 6 — ThresholdAdapter (C2) Training

209	50.4 M	0.000687	~24 min
Trainable parameters	Frozen parameters	Final MSE loss (epoch 4)	Total training time

EPOCH	AVG MSE LOSS	DURATION	CHECKPOINT
0	0.001465	5.4 min	c2_adapter_epoch_0.pth
1	0.000964	4.7 min	c2_adapter_epoch_1.pth
2	0.000947	4.6 min	c2_adapter_epoch_2.pth
3	0.000852	4.6 min	c2_adapter_epoch_3.pth
4	0.000687	4.6 min	c2_adapter_epoch_4.pth

Clamp Verification — uav0000117_02622_v (30 frames probed):

PARAMETER	VALUE	RESULT
tau_min clamp	0.300	Design parameter
tau_max clamp	0.700	Design parameter
Observed tau min	0.338972	■ Within range
Observed tau max	0.344031	■ Within range
Out-of-range	0 / 30	■ Clamp working correctly
Mean brightness	0.1805	Very dark — tau correctly near minimum

■ VERIFIED FINDING

The 61k-detection explosion on uav0000117 is NOT a clamp bug. Tau stayed within [0.3, 0.7] throughout. The issue is domain shift: at $\tau \approx 0.34$, UAV-altitude footage admits nearly all 300 det-queries per frame because MeMOTR was never trained at this altitude/scale. Fix: IOTD calibration in Phase 6 proper.

8 Evaluation Results

■ INTERIM STATUS

INTERIM RESULTS — UA-DETRAC + VisDrone only. No IOTD data. These numbers will be superseded after Phase 4 IOTD annotation.

Metric	Baseline	C2 Interim	Δ (C2–BL)	Notes
HOTA ↑	**62.13**	**60.42**	**–1.71**	**FP surge in uav0000117**
DetA ↑	55.88	52.91	**–2.97**	FP > FN reduction
AssA ↑	69.45	69.35	–0.10	■ C2 doesn't touch association
MOTA ↑	**55.19**	**47.19**	**–8.00**	**uav0000117 dominates**
IDF1 ↑	**73.57**	**70.86**	**–2.71**	
Recall ↑	77.70%	79.44%	**+1.74%**	■ More GT vehicles found
Precision ↑	77.60%	71.20%	**–6.40%**	■ UAV sequence precision hit
FP ↓	160,558	230,096	**+69,538**	58,115 from uav0000117
FN ↓	159,710	147,213	**–12,497**	■ C2 finds 12k more GT
IDSW ↓	634	819	**+185**	More tracks → more switches
MT ↑	69.73%	73.06%	**+3.33%**	■ More sequences tracked
ML ↓	8.52%	7.51%	**–1.01%**	■ Fewer sequences lost
LocA ↑	88.75	88.67	–0.08	■ Stable

9 Analysis & Interpretation

What C2 Got Right

- Recall +1.74%, FN –12,497 — C2 correctly lowers τ for dark frames, recovering genuine vehicles
- MT +3.33%, ML –1.01% — More sequences mostly tracked; fewer mostly lost
- AssA –0.10 (essentially zero) — confirms C2 only affects detection admission, not track memory
- LocA –0.08 (negligible) — bounding box quality unchanged
- Clamp verified: all tau values in [0.3, 0.7] across 30 probed frames

The uav0000117 Problem

This single UAV night sequence contributed 58,115 of the 69,538 total FP increase (84%). Mean brightness 0.1805 → τ correctly dropped to ~0.341 (near τ_min=0.3). However, at UAV altitude/resolution, MeMOTR's 300 det-queries consistently score above 0.34 even for non-vehicles —

domain shift, not a calibration error. Fix: include UAV-altitude sequences in C2 adapter training (Phase 6 proper with IOTD data).

DETRAC-Only C2 Performance: Excluding uav0000117, FP increase drops to only +11,423 with all FN/MT gains intact. Multiple DETRAC sequences show small HOTA/MOTA improvements. This validates the C2 mechanism works correctly on its training domain.

10 Roadmap — Remaining Work

PHASE	TASK	BLOCKER	PRIORITY
4	IOTD annotation export (≥3 sequences)	Manual annotation	Critical
4	Perspective calibration for C1	Phase 4 annotation	Critical
6	Retrain C2 with IOTD night/dusk data	Phase 4	Critical
6	Train C1 PositionMLP	Phase 4	High
6	Joint C1+C2 evaluation on IOTD test	C1+C2 training	High
6	Ablation table (BL vs C2 vs C1 vs C1+C2)	All runs	High
7	Final paper results (Section 12)	All eval runs	Normal

■ INTERIM DELIVERABLES COMPLETED

Phase 5/6 interim baselines complete · Raw TrackEval summaries preserved · Clamp verification done · Failure mode diagnosed · teacher_summary.md populated · All checkpoints saved (baseline epochs 0–4, C2 adapter epochs 0–4)